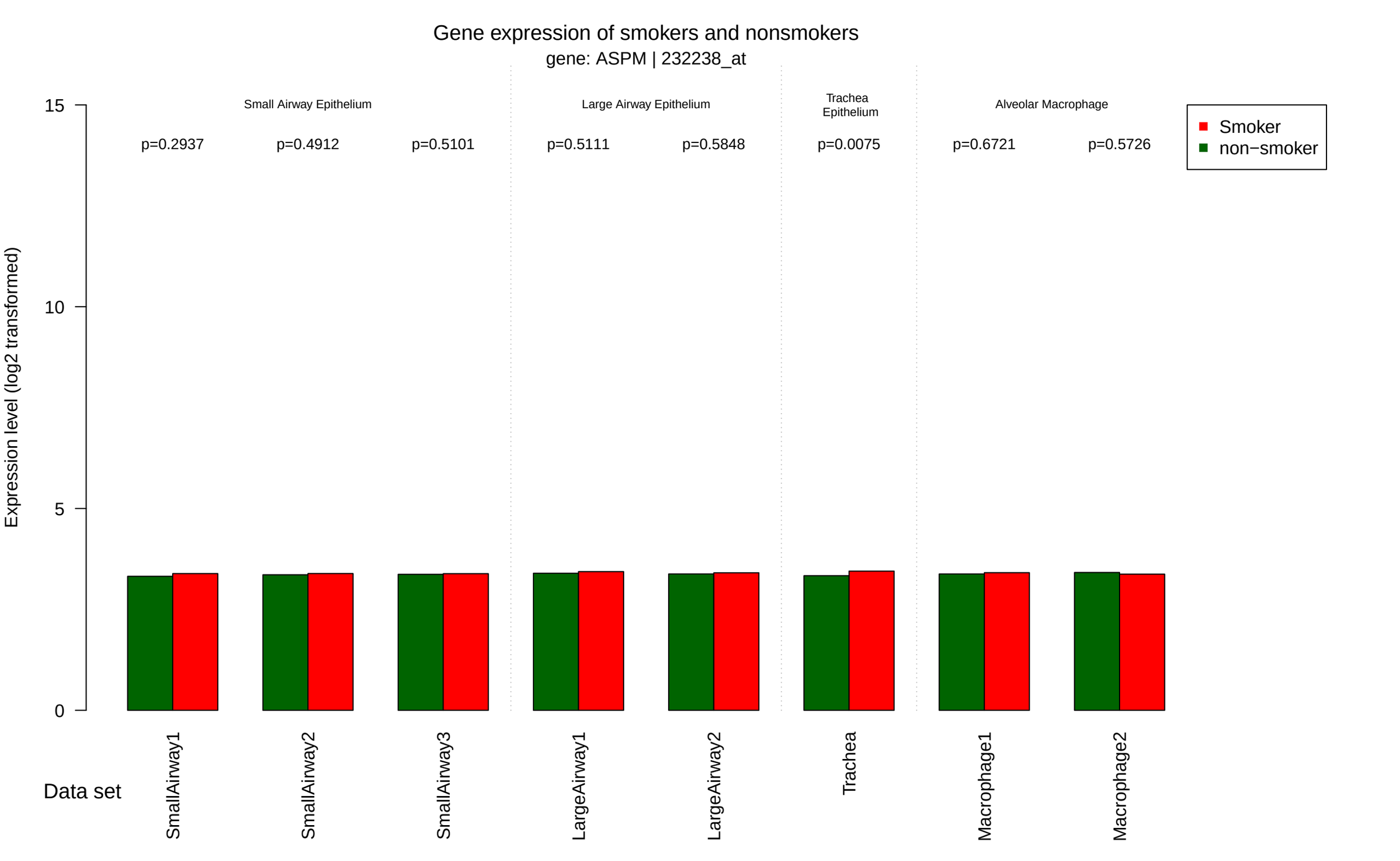
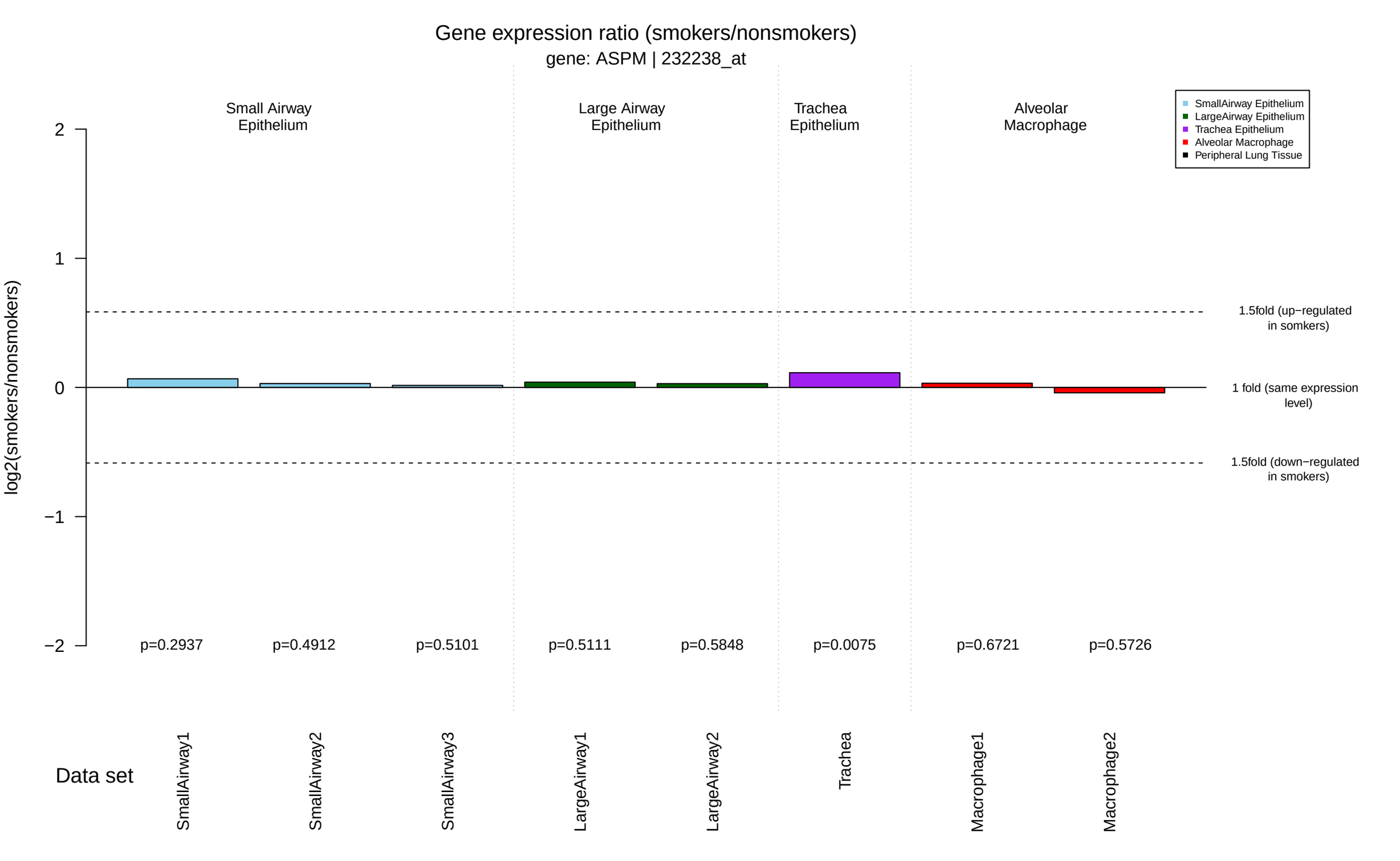
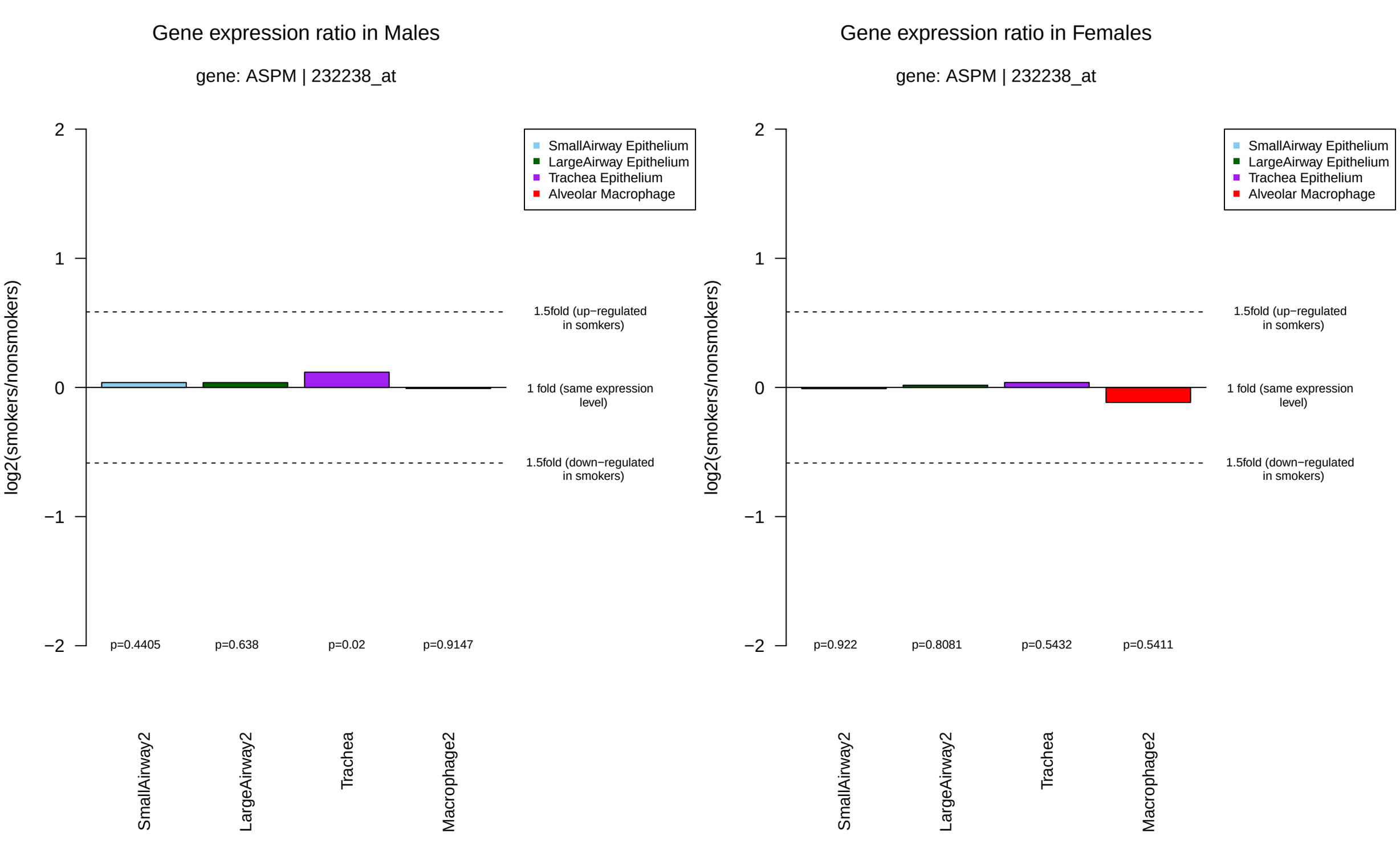
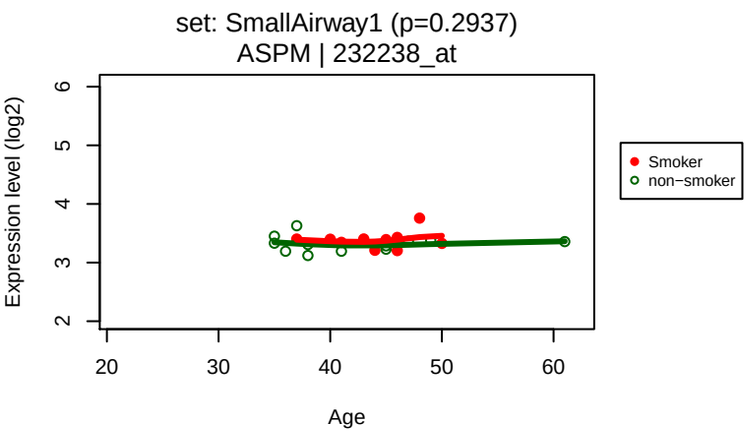
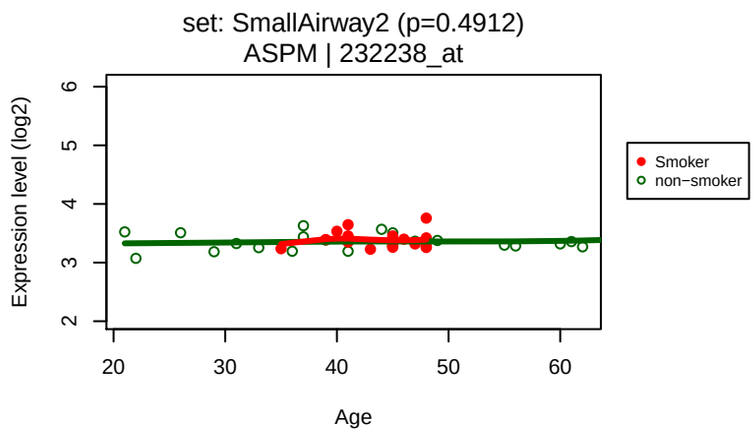
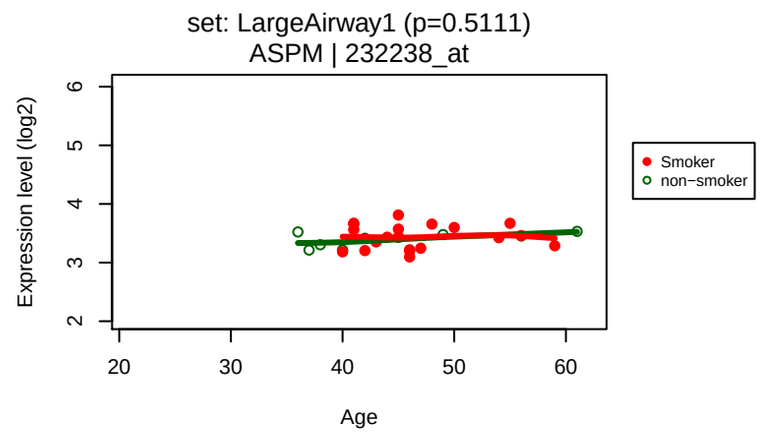

Gene expression of smokers and nonsmokers
gene: ASPM | 232238_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2937
p=0.4912
p=0.5101
p=0.5111
p=0.5848
p=0.0075
p=0.6721
p=0.5726
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

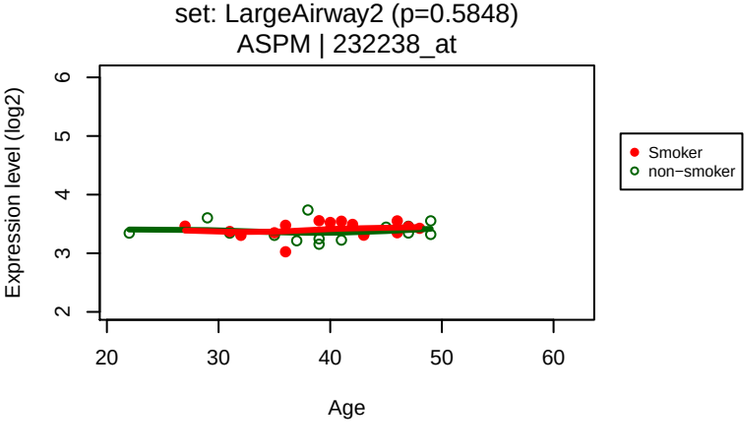

Gene expression ratio (smokers/nonsmokers)
gene: ASPM | 232238_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2937
p=0.4912
p=0.5101
p=0.5111
p=0.5848
p=0.0075
p=0.6721
p=0.5726
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

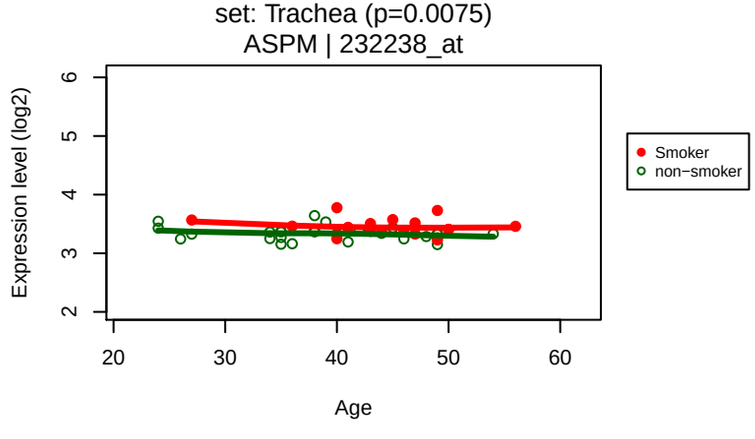

Gene expression ratio in Males
gene: ASPM | 232238_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4405
p=0.638
p=0.02
p=0.9147
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASPM | 232238_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.922
p=0.8081
p=0.5432
p=0.5411
SmallAirway2
LargeAirway2
Trachea
Macrophage2

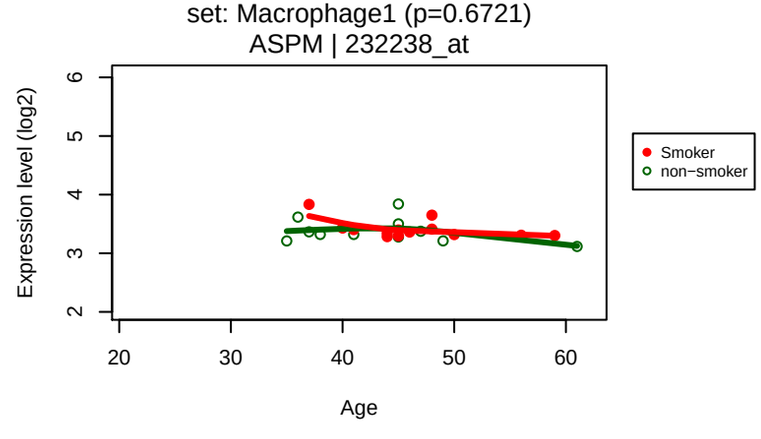

set: SmallAirway1 (p=0.2937)
ASPM | 232238_at
Expression level (log2)
Age
Smoker
non−smoker

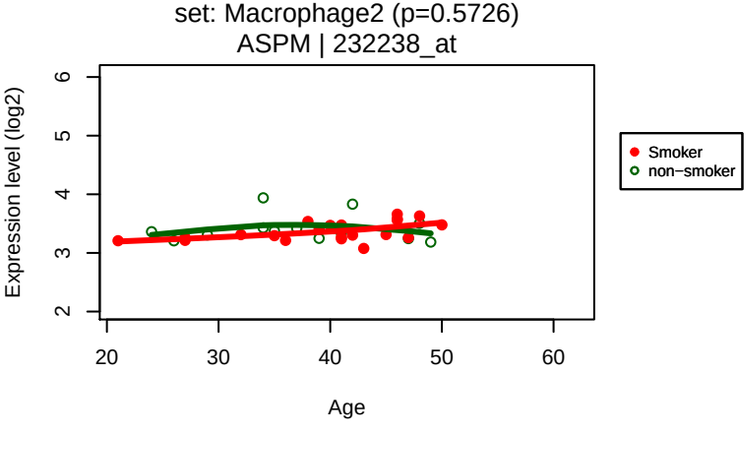

set: SmallAirway2 (p=0.4912)
ASPM | 232238_at
Expression level (log2)
Age
Smoker
non−smoker

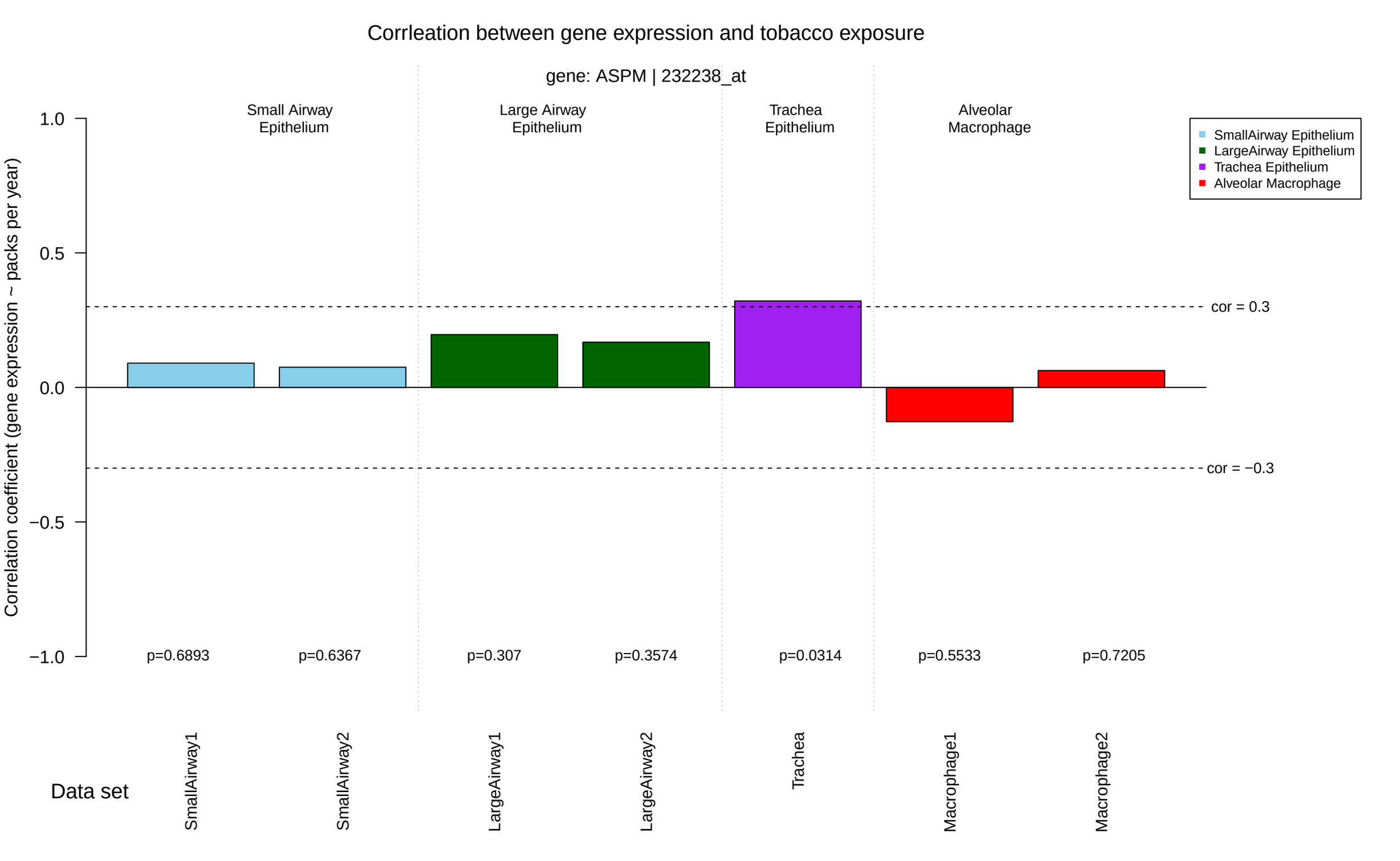

set: LargeAirway1 (p=0.5111)
ASPM | 232238_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5848)
ASPM | 232238_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0075)
ASPM | 232238_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6721)
ASPM | 232238_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5726)
ASPM | 232238_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASPM | 232238_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6893
p=0.6367
p=0.307
p=0.3574
p=0.0314
p=0.5533
p=0.7205
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2